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ESAM is a novel human hematopoietic stem cell marker associated with a subset of human leukemias

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Reliable markers are essential to increase our understanding of the biological features of human hematopoietic stem cells and to facilitate the application of hematopoietic stem cells in the field of transplantation and regenerative medicine. We previously identified endothelial cell-selective adhesion molecule (ESAM) as a novel functional marker of hematopoietic stem cells in mice. Here, we found that ESAM can also be used to purify human hematopoietic stem cells from all the currently available sources (adult bone marrow, mobilized peripheral blood, and cord blood). Multipotent colony-forming units and long-term hematopoietic-reconstituting cells in immunodeficient mice were found exclusively in the ESAM^{High} fraction of CD34⁺CD38[−] cells. The CD34⁺CD38[−] fraction of cord blood and collagenase-treated bone marrow contained cells exhibiting extremely high expression of ESAM; these cells are likely to be related to the endothelial lineage. Leukemia cell lines of erythroid and megakaryocyte origin, but not those of myeloid or lymphoid descent, were ESAM positive. However, high ESAM expression was observed in some primary acute myeloid leukemia cells. Furthermore, KG-1a myeloid leukemia cells switched from ESAM negative to ESAM positive with repeated leukemia reconstitution in vivo. Thus, ESAM is a useful marker for studying both human hematopoietic stem cells and leukemia cells. Copyright © 2016 ISEH - International Society for Experimental Hematology. Published by Elsevier Inc.

Lifelong hematopoiesis is maintained by hematopoietic stem cells (HSCs), which have the capacity for self-renewal and can differentiate into all types of blood cells. HSCs are essential for transplantation and regenerative therapies designed to cure patients with hematologic diseases. The successful outcome of therapy requires a reliable method for identifying authentic HSCs. Therefore, it is important to identify cell surface markers that distinguish HSCs from other cells.

Since a previous study indicated that a set of cell surface proteins can be used to enrich multipotent hematopoietic progenitors in the mouse bone marrow (BM) [1], numerous

studies have examined murine HSC markers (reviewed by Yokota et al. [2]). Murine HSCs can be efficiently isolated from BM as Lin[−]Sca-1⁺c-kit⁺ (LSK) CD90/Thy1^{Low}CD34^{−/Low}CD150⁺CD48[−] cells. Indeed, ~50% of cells with the LSK CD150⁺CD48[−] phenotype exhibit the potential for long-term multilineage reconstitution [3]. The same method cannot be used in humans, however, because of critical differences between the murine and human HSC phenotypes. Human HSCs express neither CD150 nor Sca-1 [4,5]. In addition, human HSCs are enriched in the CD34⁺CD38[−] fraction [6,7], whereas many studies have located murine HSCs in the CD34[−]CD38⁺ fraction [8–10]. These discrepancies have limited the translation of findings from mice to humans. Therefore, the identification of HSC antigens common to the two species would be a significant advancement in translational studies of HSC biology.

We previously identified endothelial cell-selective adhesion molecule (ESAM) as a novel marker for murine HSCs

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[11]. The functional importance of ESAM is underscored by the finding that ESAM-deficient mice suffer from severe and prolonged BM suppression after treatment with the cytotoxic agent 5-fluorouracil (5-FU) [12]. The detection of ESAM transcripts in human cord blood (CB) $CD34^+CD38^-Lin^-CD90/Thy1^+$ cells has increased interest in ESAM as an HSC marker [13]. However, it is unclear whether its expression pattern changes in human hematopoietic stem/progenitor cells with the differentiation status, or in a manner that depends on the tissue source. In addition, the expression of ESAM in a broad panel of human leukemia cells has not been evaluated.

In this study, we examined whether ESAM expression is a potential marker of human HSCs from diverse sources. Interestingly, in addition to adult BM (ABM), most $CD34^+CD38^-$ cells in granulocyte colony-stimulating factor (G-CSF)-mobilized peripheral blood (GMPB) expressed ESAM. We also identified a previously uncharacterized subset of CB $CD34^+CD38^-$ cells that expressed extremely high levels of ESAM. Furthermore, we observed oscillations in ESAM expression on leukemia cells depending on the surrounding environment.

Methods

Bone marrow and GMPB were obtained from related HSC transplantation donors. CB was obtained from healthy full-term newborns. BM was also obtained from patients with acute leukemia and the femoral heads of patients who underwent hip replacement surgery. The institutional review board at Osaka University Hospital approved all protocols, and written informed consent was obtained from all participants. To prepare BM cells adjacent to bone tissues, trabecular tissues of the femora were treated with 2 mg/mL collagenase IV and DNase and gently agitated for 1 hour at 37°C.

Additional methods are described in detail in the [Supplementary Methods](#) (online only, available at www.exphem.org).

Results

High ESAM expression indicates primitive hematopoietic progenitors in human $CD34^+CD38^-$ cells

Our previous study indicated that sorting for high ESAM expression enriches the proportion of authentic HSCs in the LSK fraction of murine fetal liver and ABM. In this study, we examined whether this was also true for human cells. Using flow cytometry, we analyzed ESAM expression on BM mononuclear cells (MNCs). A large number of human BM $CD34^+$ cells were found to express ESAM, and most $CD34^+CD38^-$ cells, which included the human HSC-enriched fraction, expressed ESAM (Fig. 1A). We found that in both ABM and GMPB, more than 80% of $CD34^+CD38^-$ cells expressed high levels of ESAM (Fig. 1B, C). The addition of anti-CD90 and

CD45RA antibodies can reportedly divide the $CD34^+CD38^-$ fraction into three subpopulations, including $CD90^+CD45RA^-$ HSCs, $CD90^-CD45RA^-$ multipotent progenitors, and $CD90^-CD45RA^+$ multilymphoid progenitors (MLPs) [14] (Fig. 1D, left). We found that most HSCs expressed high levels of ESAM, whereas many MLPs lost ESAM expression (Fig. 1D, middle). Furthermore, estimation of the mean fluorescence intensity indicated that ESAM expression markedly declined as the cells progressively differentiated from HSCs into MLPs (Fig. 1D, right).

Next, we compared the differentiation and growth potential of $ESAM^{-/Low}$ and $ESAM^{High}$ cells. We purified $CD34^+CD38^-ESAM^{-/Low}$ and $CD34^+CD38^-ESAM^{High}$ cells from ABM, and subjected the cells to a methylcellulose assay or grew the cells in MS5 stromal cell co-culture. Although both cell types produced hematopoietic colonies with high frequency, only $ESAM^{High}$ cells included lineage-mixed colony-forming units (CFU-Mix) with both myeloid and erythroid potential (Fig. 2A). Under the MS5 co-culture conditions, the $ESAM^{-/Low}$ cells generated $CD45^+$ hematopoietic cells more effectively than the $ESAM^{High}$ cells during the first week (Fig. 2B, left); however, the output of $CD45^+$ cells did not increase to the same extent as that from $ESAM^{High}$ cells during the second week and thereafter (Fig. 2B, right). Indeed, hematopoietic cells rapidly expanded from the $ESAM^{High}$ fraction after the second week, and this was maintained throughout the culture period (Fig. 2B). In addition, the $ESAM^{High}$ fraction gave rise to a substantial number of $CD19^+$ B-lineage cells (Fig. 2C). Similar results were obtained when GMPB-derived cells were used (data not shown). These observations indicate that although both the $ESAM^{-/Low}$ and $ESAM^{High}$ cells in $CD34^+CD38^-$ included hematopoietic progenitors, higher ESAM expression distinguished the more primitive and multipotent cells.

High ESAM expression reveals a new subpopulation of human CB $CD34^+CD38^-$ cells

We also analyzed ESAM expression on human CB MNCs. Similar to ABM and GMPB, most cells in the CB $CD34^+CD38^-$ fraction expressed ESAM. However, we detected a new subfraction of cells exhibiting very high ESAM expression, referred to as $ESAM^{Bright}$, which was not observed in the ABM and GMPB $CD34^+CD38^-$ fractions. The $ESAM^{Bright}$ fraction had approximately 10-fold higher expression of ESAM as compared with the $ESAM^{High}$ fraction. On the basis of ESAM expression level, we subdivided the CB $CD34^+CD38^-$ fraction into three populations: $ESAM^{-/Low}$, $ESAM^{High}$, and $ESAM^{Bright}$ (Fig. 3A). Although the percentage of $ESAM^{-/Low}$ cells among the total $CD34^+CD38^-$ cells was consistently low in the CB samples tested, the frequencies of $ESAM^{High}$ and $ESAM^{Bright}$ cells varied widely among samples (Fig. 3B).

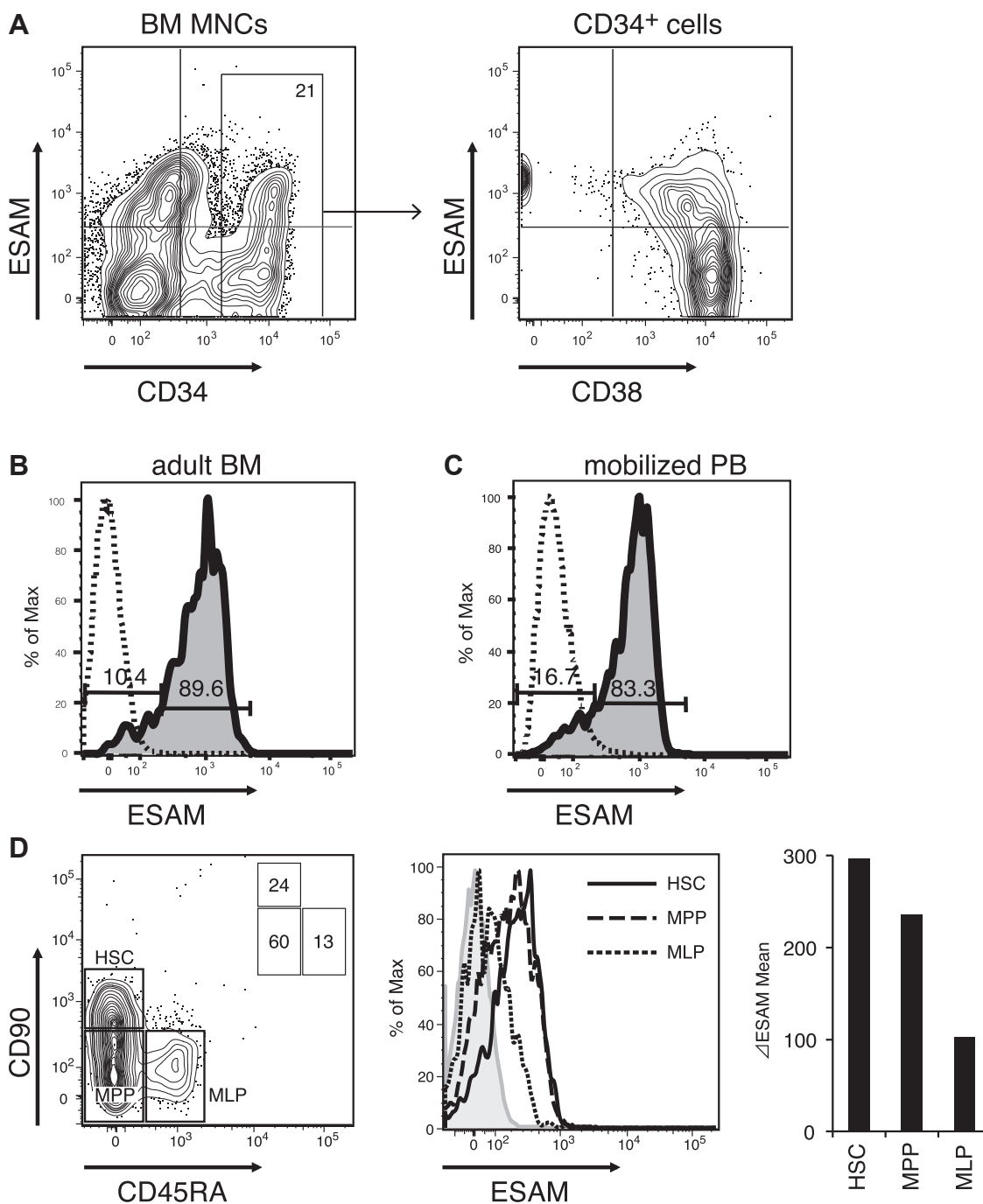


Figure 1. Expression of ESAM on human BM and GMPB. (A) Mononuclear cells (MNCs) from human BM were stained with anti-CD34, anti-CD38, and anti-ESAM antibodies and analyzed by flow cytometry. (B, C) ESAM expression patterns of the CD34⁺CD38⁻ fractions of BM (B) and GMPB (C). *Solid black lines* and *dashed lines* represent ESAM and background levels, respectively. The CD34⁺CD38⁻ fraction was subdivided into two populations, ESAM^{-/Low} and ESAM^{High}, and the percentages of cells in each gate are shown in each panel. (D) Left: CD34⁺CD38⁻ cells in ABM were stained with CD45RA and CD90 (Thy-1) and subdivided into three fractions: CD45RA⁻CD90⁺ hematopoietic stem cells (HSC); CD45RA⁻CD90⁻ multipotent progenitors (MPP); and CD45RA⁺CD90⁻ multilymphoid progenitors (MLP). The percentage of cells in each fraction is provided in the upper right. Middle: ESAM expression levels in HSCs (*solid line*), multipotent progenitors (*dashed line*), and MLPs (*dotted line*) are shown. The *gray line* represents the background level of CD34⁺CD38⁻ cells. Right: Differences between background and mean ESAM intensities in each fraction.

ESAM^{High}, but not ESAM^{Bright}, indicates authentic HSCs in CB

Next, we performed functional assessment of the subpopulations based on ESAM expression levels. In some cases,

ESAM^{-/Low} cells were omitted from the assessment because their frequency in the CB CD34⁺CD38⁻ fraction was too low for accurate evaluation. We predicted that ESAM^{Bright} cells would be enriched for more primitive

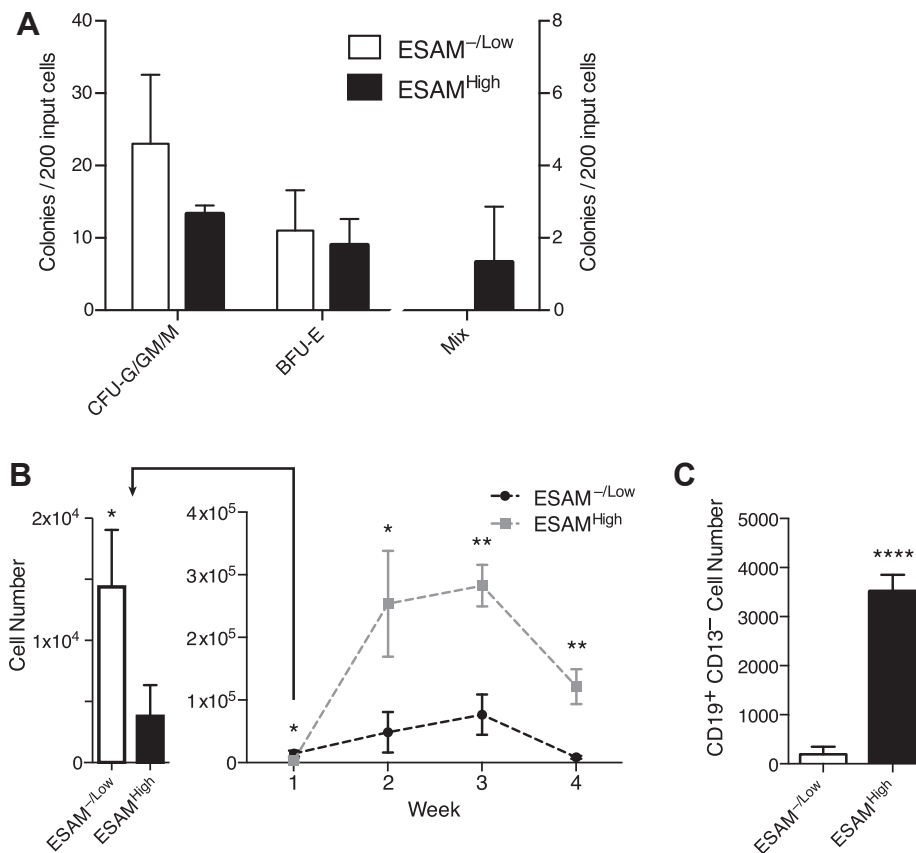


Figure 2. High ESAM expression marks primitive hematopoietic progenitors. (A) ESAM^{-Low} and ESAM^{High} cells of the CD34⁺CD38⁻ fraction of BM were sorted and subjected to a methylcellulose colony formation assay. Numbers of colony-forming units (CFUs) are shown. CFU-G/GM/M = CFU granulocyte/granulocyte-macrophage/macrophage; BFU-E = burst-forming unit erythrocyte; CFU Mix = mixed colony-forming unit. (B) ESAM^{-Low} and ESAM^{High} cells of the CD34⁺CD38⁻ fraction of BM were sorted and co-cultured with MS5 in the presence of stem cell factor, Flt3 ligand, and interleukin-7. Cell numbers of each fraction after 1–4 weeks are shown. **p* < 0.05, ***p* < 0.005 by Student's *t* test. (C) Numbers of CD19⁺ CD13⁻ B-lineage cells after 4-week co-culture with MS5. *****p* < 0.0001 by Student's *t* test.

progenitors; however, the opposite was observed. In methylcellulose assays, the ESAM^{High} fraction produced significantly more CFUs than the ESAM^{Bright} fraction (Fig. 4A). Furthermore, only the ESAM^{High} fraction was capable of colony formation in the CFU-Mix, which contained primitive multipotent progenitors.

To determine the frequency of primitive progenitors with both lymphoid and myeloid potential, limiting dilution analyses were performed in the MS5 co-culture system. Although 1 in 12 ESAM^{High} cells produced both CD19⁺ lymphoid-lineage and CD33⁺ myeloid-lineage cells, only 1 in 63 ESAM^{Bright} cells produced cells of both lineages (Fig. 4B). These results suggest that primitive myelolymphoid progenitors are enriched in the ESAM^{High} fraction, but not in the ESAM^{Bright} population, of human CB CD34⁺CD38⁻ cells.

To evaluate long-term reconstitution activity, we first transplanted ESAM^{High} or ESAM^{Bright} cells from the CB CD34⁺CD38⁻ fraction into 2.2-Gy-irradiated NOD/Shi-scid, IL-2γ-null (NOG) mice. While all five recipients of the ESAM^{Bright} cells died within 1 month after transplantation, all five recipients of the ESAM^{High} cells survived.

Surviving ESAM^{High}-transplanted mice were sacrificed 7 months after transplantation, and BM was evaluated. We found that the ESAM^{High} cells effectively reconstituted human hematopoiesis in all recipients (data not shown). Next, we performed serial transplantation assays to compare ESAM^{-Low}, ESAM^{High}, and ESAM^{Bright} cells purified from the CB CD34⁺CD38⁻ fraction. The number of transplanted cells in each group was determined according to their frequency in the CB CD34⁺CD38⁻ fraction (see Fig. 4 legend). Because all ESAM^{Bright}-transplanted mice died before assessment in the first experiment, we reduced the irradiation dose to 1.8 Gy. All recipients survived and were analyzed 3 months after transplantation. Human hematopoiesis was observed only in mice transplanted with ESAM^{High} cells (Fig. 4C). Furthermore, transplanted cells contributed substantially to the myeloid, B-lymphoid, and T-lymphoid lineages (Fig. 4D). In the secondary transplantation, whole BM MNCs from the recipients of ESAM^{High} cells were transplanted into sublethally irradiated NOG mice. All mice were sacrificed 4 months after transplantation, and chimerism of human cells in the BM was evaluated. Long-term human

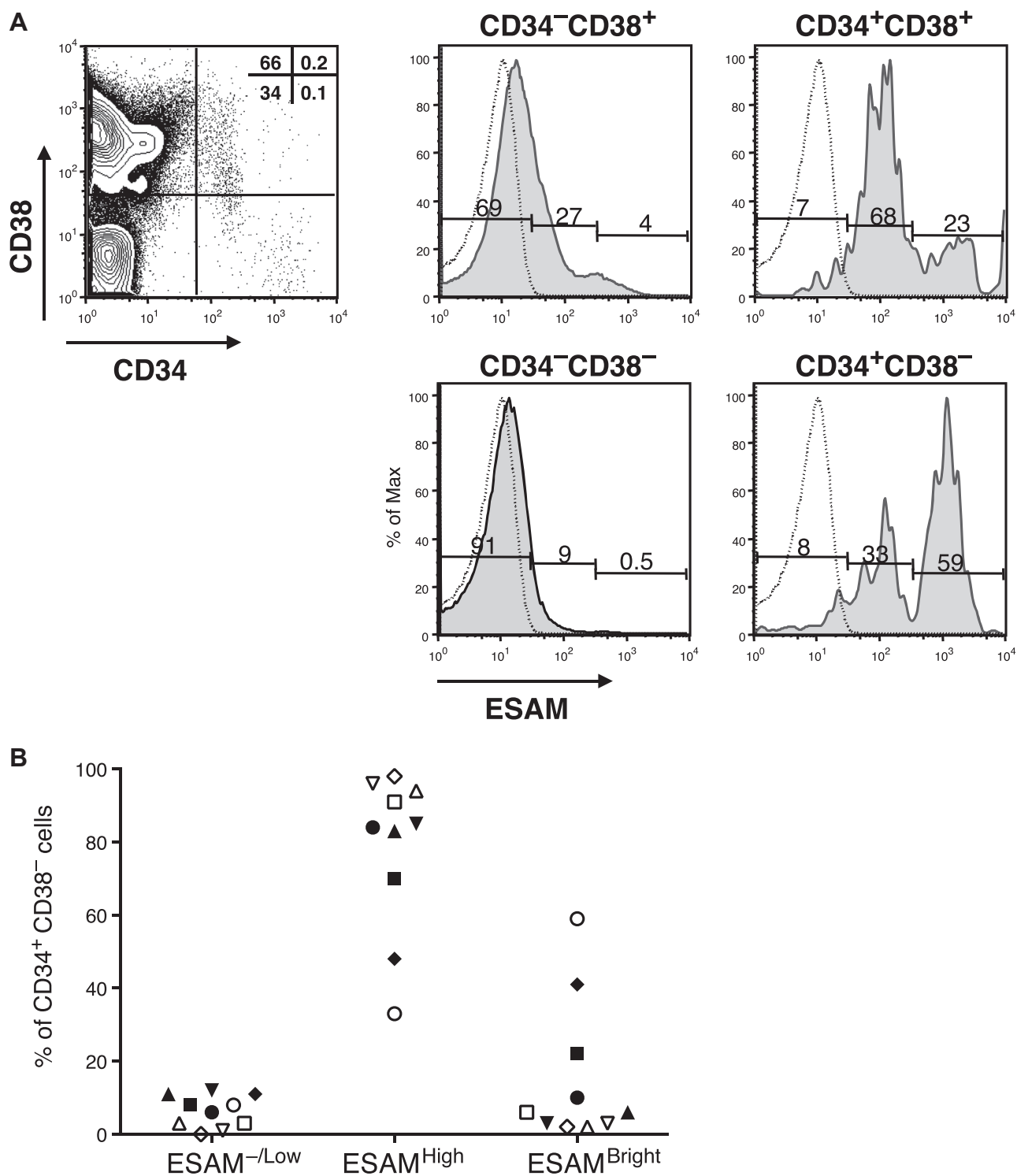


Figure 3. Expression of ESAM on human CB cells. **(A)** Left: CB MNCs were stained with anti-CD34 and -CD38 antibodies and analyzed by flow cytometry. Percentages of cells in each subpopulation are at the upper right. Right: ESAM expression in CD34⁻CD38⁺, CD34⁺CD38⁺, CD34⁻CD38⁻, and CD34⁺CD38⁻ fractions was analyzed by flow cytometry. According to ESAM expression levels, the CD34⁺CD38⁻ fraction was subdivided into three populations: ESAM^{-/Low}, ESAM^{High}, and ESAM^{Bright} (lower right). The same gates were applied to other fractions, and the percentage of cells in each gate is given in each panel. *Solid black lines* and *dashed lines* represent ESAM and background levels, respectively. **(B)** Ten CB samples were analyzed in the same manner, and the percentages of ESAM^{-/Low}, ESAM^{High}, and ESAM^{Bright} cells are provided. The percentage of each population of the same CB sample is represented by the same symbols.

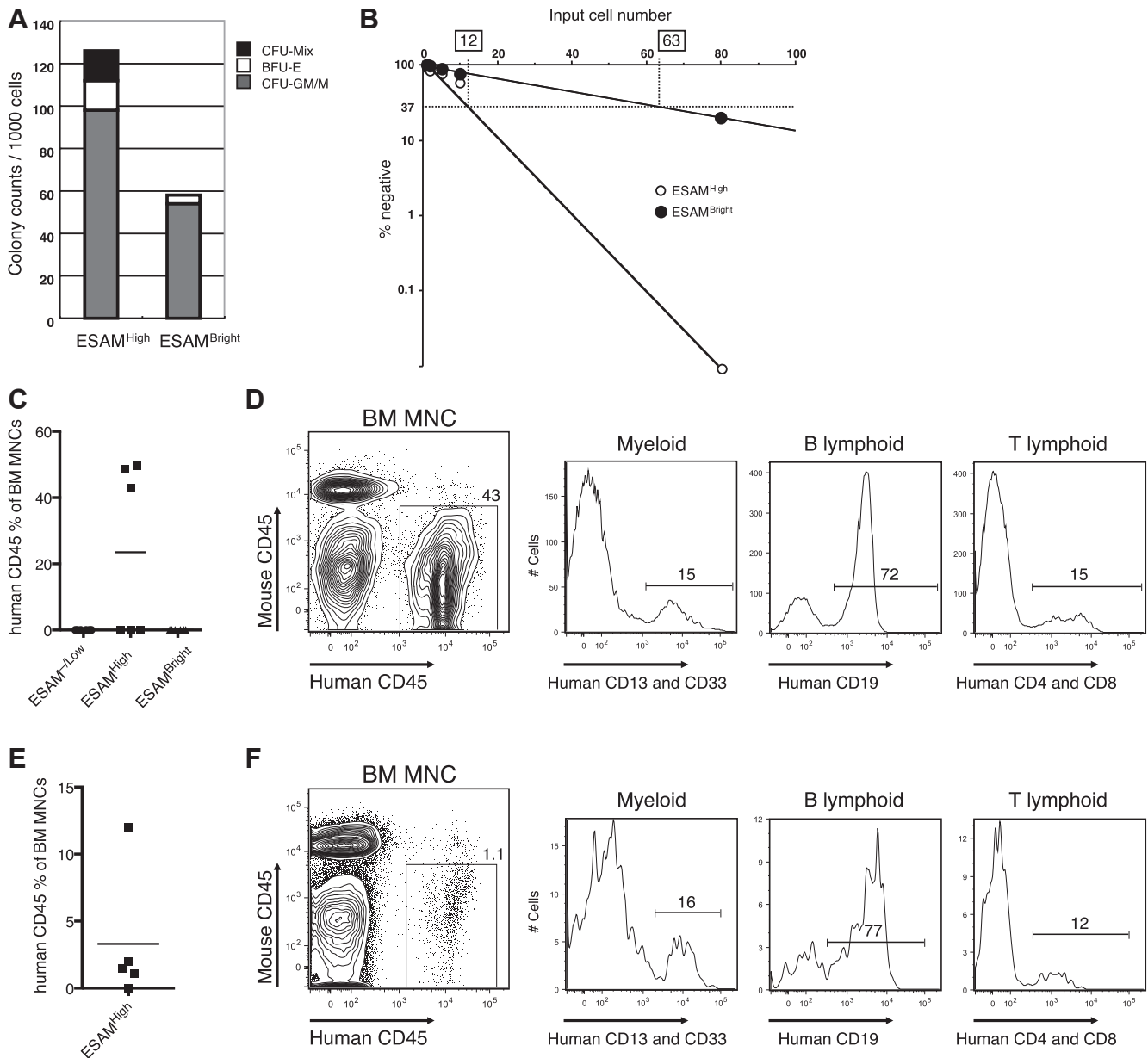


Figure 4. High ESAM expression marks authentic HSCs in CB. (A) ESAM^{High} and ESAM^{Bright} cells in the CB CD34⁺CD38⁻ fraction were sorted and subjected to methylcellulose colony formation assay. Numbers of colony-forming units (CFUs) are shown. BFU-E = burst-forming unit erythrocyte; CFU-GM/M = colony-forming unit granulocyte–macrophage/macrophage; CFU-Mix = mixed colony-forming unit. (B) ESAM^{High} and ESAM^{Bright} cells of CB CD34⁺CD38⁻ fraction were subjected to limiting dilution analyses in the MS5 co-culture system in the presence of stem cell factor and granulocyte colony-stimulating factor (G-CSF). The input cell numbers corresponding to 37% negative value are in rectangles. (C) ESAM^{Low}, ESAM^{High}, and ESAM^{Bright} cells in the CB CD34⁺CD38⁻ fraction were purified and transplanted into sublethally irradiated NOG mice (ESAM^{Low} 5×10^3 cells/mouse, ESAM^{High} 2×10^4 cells/mouse, ESAM^{Bright} 1×10^3 cells/mouse, $n = 6$ for each group). Three months after transplantation, all recipients were sacrificed, and flow cytometry analyses of BM were performed. BM MNCs were stained with mouse CD45 and human CD45, and the percentages of human CD45⁺ cells are shown. (D) Representative results of the recipient transplanted ESAM^{High} cells. Human CD45⁺ cells were gated, and the percentages of each lineage marker-positive cell are shown. (E) Whole BM MNCs were obtained from primary recipients of ESAM^{High} cells and were transplanted into sublethally irradiated NOG mice (2×10^6 cells/mouse, $n = 5$). Four months after transplantation, all recipients were sacrificed, and flow cytometry analyses of BM were performed. Percentages of human CD45⁺ cells are shown. (F) Representative results for the secondary recipient.

hematopoiesis was again detected in the three lineages (Fig. 4E, 4F). Thus, the ESAM^{High} fraction, but not the ESAM^{Low} and ESAM^{Bright} fractions, contains long-term reconstituting human HSCs.

CD34⁺CD38⁻ ESAM^{Bright} fraction consists of non-hematopoietic-lineage cells

The unexpected results of our functional evaluation of CB CD34⁺CD38⁻ ESAM^{Bright} cells prompted us to

characterize these cells more precisely. Morphologically, ESAM^{Bright} cells were larger and had less basophilic and more abundant cytoplasm compared with ESAM^{Low} and ESAM^{High} cells (Fig. 5A). Although ESAM^{Low} and ESAM^{High} cells were of uniform size and morphology, ESAM^{Bright} cells were polymorphic, suggesting that they are heterogeneous.

With respect to the cell surface phenotype, ESAM^{Bright} cells did not express CD45 (Fig. 5B, left). In addition, although ESAM^{High} cells expressed CD133, another marker of human HSCs [15,16], this marker was completely absent in ESAM^{Bright} cells (Fig. 5B, right). Back-plotting of the ESAM^{High} and ESAM^{Bright} cells onto the CD34 and CD38 profiles revealed that CD34 expression levels were much higher on ESAM^{Bright} cells (Fig. 5C). Furthermore, these cells expressed vascular endothelial (VE)-cadherin and vascular endothelial growth factor receptor 2 (VEGFR-2), which are endothelial lineage-related molecules (Fig. 5D). ESAM^{Bright} cells also expressed CD146/S-end, and a small subset was positive for the expression of CD118/leukemia inhibitory factor receptor (data not shown). These results suggest that the main population of CB CD34⁺CD38[−] ESAM^{Bright} cells was not hematopoietic, but rather, was composed of endothelial-lineage and other mesodermal cells.

On the basis of the aforementioned findings, we assumed that a similar ESAM^{Bright} cell population existed in BM. Because aspirated BM cells did not contain ESAM^{Bright} cells (Fig. 1B), we treated bone samples with collagenase to collect mesenchymal cells that are adherent to bone tissues. This treatment yielded a substantial CD34⁺CD38[−] ESAM^{Bright} population from ABM (Fig. 5E), the phenotype of which was similar to that of its CB counterpart. We then performed a microarray experiment to compare the gene expression profiles of the CD34⁺CD38[−]ESAM^{High} and CD34⁺CD38[−]ESAM^{Bright} populations. The former cells expressed HSC-related genes, whereas the latter had more endothelial-related profiles [17] (Fig. 5F) (the data have been deposited in Gene Expression Omnibus under Accession No. GSE63877). Indeed, CD34⁺CD38[−] ESAM^{Bright} cells from collagenase-treated bone produced CD31⁺ cells, but not CD45⁺ hematopoietic cells, in co-culture with MS5 stromal cells in the presence of VEGF, stromal cell-derived factor, and interleukin (IL)-16 (Fig. 5G). These observations indicate that the CD34⁺CD38[−] fraction, which is conventionally considered the human HSC fraction, may also contain a substantial number of non-hematopoietic cells. Thus, the inclusion of ESAM or other markers appears to provide a more accurate estimation of HSC numbers.

Some types of human leukemia cells express ESAM

Cell surface antigens on normal hematopoietic stem/progenitor cells are often observed on leukemia cells. Because some of these antigens are useful for determining leukemia

lineage and have utility as prognostic indicators [18,19], we examined whether ESAM would be a valuable addition to this antigen panel. First, we examined human leukemia cell lines derived from patients with acute myeloid leukemia (AML); all of these cell lines were uniformly negative for ESAM expression (Fig. 6A). KG-1a cells exhibited a contrasting profile for CD34 and ESAM. Lymphoid-lineage cell lines were also negative for ESAM. In contrast, the erythroid leukemia cell line HEL and the megakaryocytic leukemia cell line CMK had high ESAM expression. Additionally, K562 cells that originated from a chronic myeloid leukemia that subsequently transformed into acute erythroleukemia also expressed low levels of ESAM. These results indicate that ESAM expression segregates leukemia cells of erythroid or megakaryocyte lineage.

We then evaluated ESAM expression on primary leukemia cells isolated from patients with AML ($n = 15$) and acute lymphoblastic leukemia (ALL, $n = 3$) at the time of diagnosis. Interestingly, although ALL cases were nearly negative for ESAM, 10 of 15 AML cases were clearly ESAM positive (Fig. 6B). Notably, the ESAM expression pattern was completely different in the same AML category (Fig. 6B). Thus, AML cells may change their ESAM expression levels according to cell-intrinsic features and/or the surrounding environment.

KG-1a leukemic cells express ESAM concomitant with the acquisition of a more aggressive phenotype

The discrepancy between AML cell lines and primary AML cells with respect to ESAM expression indicates that this antigen is differentially regulated in vivo. Therefore, we inoculated ESAM[−] KG-1a cell lines into NOD/SCID mice and harvested reconstituted KG-1a (rKG-1a) cells 10 weeks after inoculation. The cells were then cultured in vitro for 3 weeks and inoculated again into NOD/SCID mice (Fig. 7A). These rKG-1a cells also exhibited high leukemic activity (Fig. 7B). Flow cytometry analyses revealed that although parental KG-1a cells were ESAM negative, rKG-1a cells exhibited high ESAM expression (Fig. 7C). To investigate the significance of ESAM upregulation in rKG-1a cells, ESAM[−] and ESAM⁺ rKG-1a cells were purified and inoculated into sublethally irradiated NOD/SCID mice. Three months after transplantation, all mice were sacrificed, and the BM cells were analyzed. Flow cytometry analyses revealed that ESAM[−] rKG-1a cells did not engraft, whereas ESAM⁺ rKG-1a cells engrafted (Fig. 7D, E). These results indicate that leukemia cells fluctuate in their surface phenotype according to the surrounding environment and that ESAM expression is related to the affinity of leukemia cells to BM stromal cells.

Discussion

Precisely defining the immunophenotype of HSCs is critical to the successful outcome of transplantation therapy for

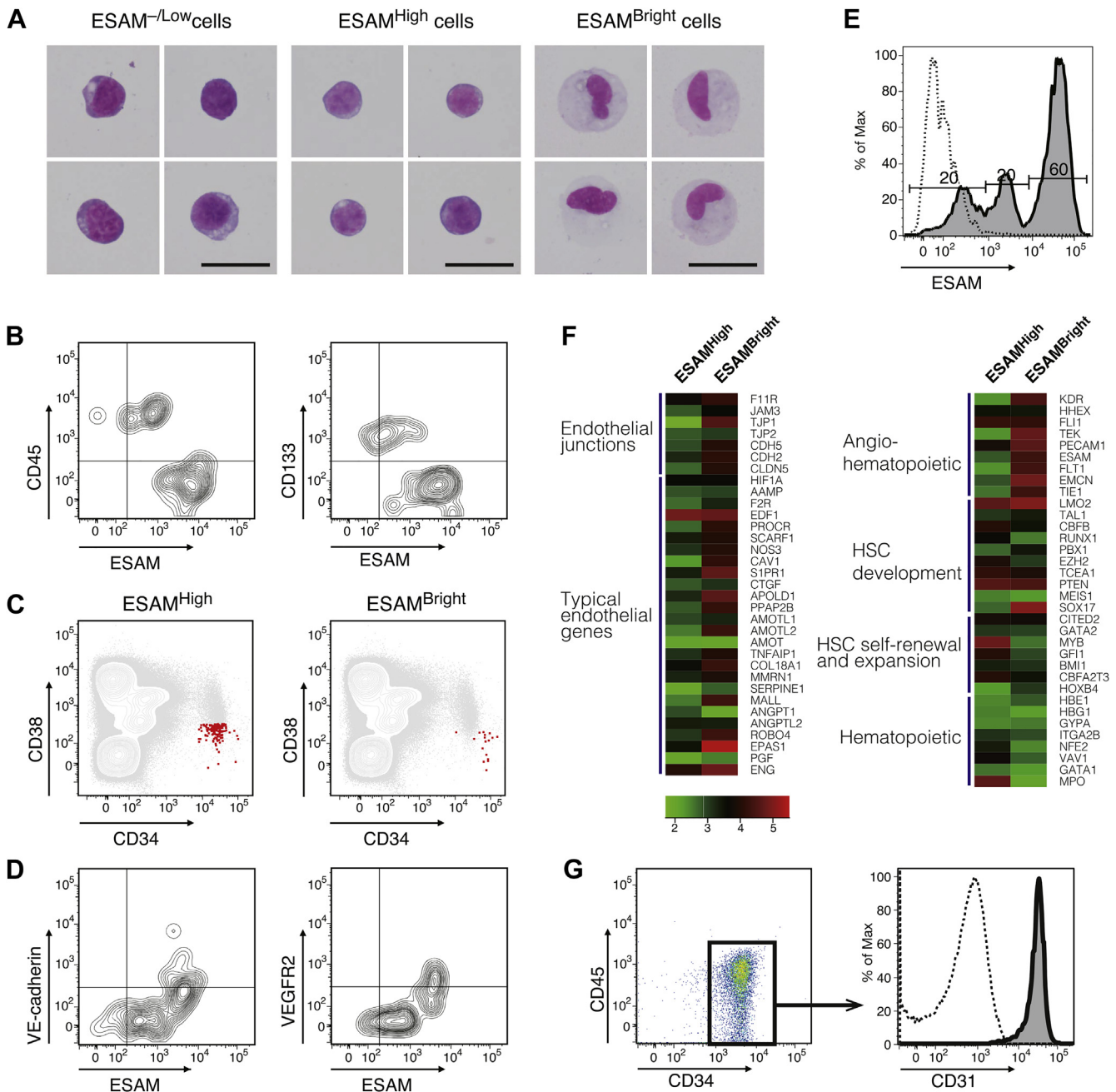


Figure 5. Morphologic, immunophenotypic, and gene expression profiling differences between ESAM^{High} and ESAM^{Bright} cells of the CD34⁺CD38⁻ fraction. (A) ESAM^{Low}, ESAM^{High} and ESAM^{Bright} cells of CB CD34⁺CD38⁻ fraction were sorted, and cytospin slides were stained by May–Grünwald/Giemsa staining. Horizontal bars represent 20 μ m. (B) CB CD34⁺CD38⁻ cells were stained with ESAM and CD45, hematopoietic cell marker (left), and CD133, HSC marker (right). (C) ESAM^{High} and ESAM^{Bright} cells of the CB CD34⁺CD38⁻ fraction were back-plotted onto the CD34 and CD38 profile. ESAM^{High} cells were found as CD34^{Low}, whereas ESAM^{Bright} cells were CD34^{High}. (D) CB CD34⁺CD38⁻ cells were stained with ESAM and the endothelial markers, VEGFR-2 (left panel) and VE-cadherin (right panel). (E) ESAM expression of the CD34⁺CD38⁻ fraction from collagenase-treated bone was determined by flow cytometry. ESAM^{Low}, ESAM^{High}, and ESAM^{Bright} gates and the percentages of cells in each gate are shown. The solid black line and dashed line represent ESAM and background levels, respectively. (F) Gene expression profiles of CD34⁺CD38⁻ ESAM^{High} and ESAM^{Bright} cells from collagenase-treated bone were determined by microarray. Heat maps of selected genes related to endothelial and hematopoietic cells are shown. (G) CD34⁺CD38⁻ ESAM^{Bright} cells from collagenase-treated bone were cultured with MS5 stromal cells in the presence of VEGF, stromal cell-derived factor (SDF), and IL-16. This co-culture condition supports growth of both endothelial and hematopoietic cells simultaneously [41]. Left: After 5 weeks, the cells were harvested and analyzed by flow cytometry. Recovered cells were stained with anti-CD34 and -CD45 antibodies. Right: Expression of endothelial marker CD31 on the CD34⁺CD45⁻ cells.

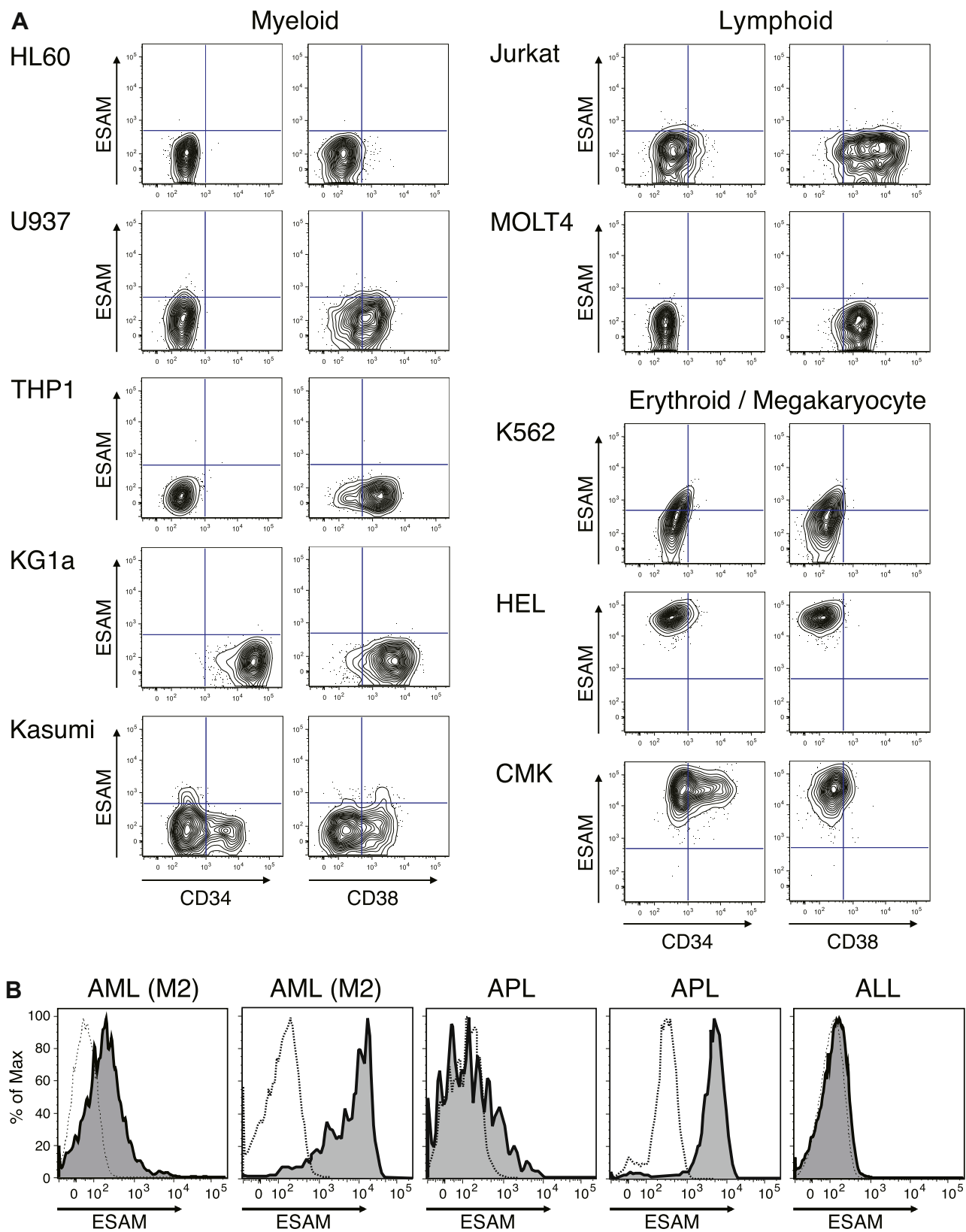


Figure 6. ESAM expression on various human leukemia cell lines and primary leukemia cells. **(A)** ESAM along with CD34 and CD38 expression patterns on various human leukemia cell lines were analyzed by flow cytometry. **(B)** Primary leukemia cells were analyzed by flow cytometry. Leukemia cells were gated in the CD34 and CD38 profiles, and the ESAM expression patterns of leukemia cells are shown. Solid black lines and dashed lines represent ESAM and background levels, respectively.

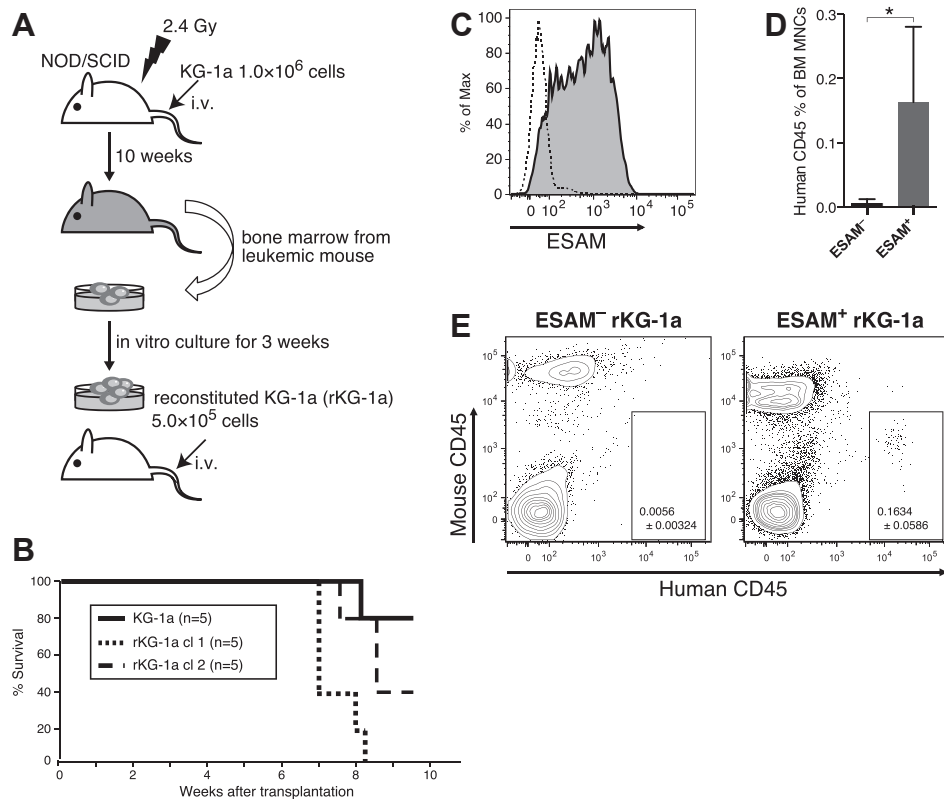


Figure 7. KG-1a leukemic mouse model. (A) Scheme of the transplantation protocols. KG-1a, a human AML cell line, was transplanted into sublethally irradiated NOD/SCID mice. Ten weeks after transplantation, the mice were killed, and leukemic cells in BM were collected. The reconstituted KG-1a (rKG-1a) was cultured in vitro for 3 weeks and re-transplanted into sublethally irradiated NOD/SCID mice. (B) Kaplan–Meier survival curves of transplanted mice. (C) ESAM expression patterns of KG-1a and rKG-1a were analyzed by flow cytometry using a monoclonal antibody specific for human ESAM. The solid black line and dashed line represent ESAM expression levels in rKG-1a and KG-1a, respectively. (D) ESAM⁺ and ESAM⁻ rKG-1a cells were sorted and transplanted into sublethally irradiated NOD/SCID mice (5×10^4 cells/mouse, $n = 4$ for each group). Three months after transplantation, the mice were sacrificed, and BM cells were analyzed by flow cytometry. Percentages of human CD45⁺ cells are shown. * $p < 0.05$ by Student's *t*-test. (E) Representative results of the mice transplanted with ESAM⁻ and ESAM⁺ rKG-1a cells. Percentages of human CD45⁺ cells (mean $\pm$ SD) are in the rectangles.

hematologic malignancies. In addition, recent progress in regenerative therapy using induced pluripotent stem cells demands that reliable cell surface antigens are available to distinguish authentic HSCs from lineage-specific progenitors [20,21]. However, it is more difficult to characterize human HSCs and their proximate progenitors compared with murine HSCs, in part because of the limited information available regarding human HSC surface antigens [2,22]. The developmental process of HSCs with respect to ontogeny and transcription factors is thought to be similar in mice and humans. Therefore, identifying common HSC antigens shared by the two species will greatly facilitate basic and clinical studies of HSC biology. In this study, we found that ESAM is a robust and cross-species HSC marker. Furthermore, the molecule is functional on HSCs and is associated with specific subsets of leukemia cells.

Using strict phenotypic definitions of human HSCs and their proximate progenitors in the CD34⁺CD38⁻ fraction [14], we found that the CD90⁺CD45RA⁻ HSCs expressed high levels of ESAM and that this expression decreased

during their differentiation into lymphoid progenitors. In our culture and transplantation experiments, we also found that high ESAM expression segregates with multilineage-potential and SCID-repopulating HSCs in humans; these data are consistent with our previous findings in mice [11]. In addition, and similar to the case for murine HSCs, ESAM was evidently a robust marker of human HSCs regardless of changes to their functional properties and expression of other cell surface antigens. We also assessed the long-term reconstitution capacity of ESAM^{High}CD34⁺CD38⁻ cells from various tissues (BM, GMPB, and CB) simultaneously and observed comparable results (data not shown).

Our previous study highlighted the functional importance of ESAM in the proliferation and differentiation of HSCs. ESAM expression is markedly increased on murine HSCs following cytotoxic insult [12]. Additionally, ESAM-deficient mice are less tolerant to 5-fluorouracil treatment, and the recovery of their HSC fraction was significantly delayed compared with that of wild-type mice [12]. Furthermore, we found that ESAM is vital for the early stages of

erythrocyte recovery after BM injury (T. Sudo, manuscript in preparation). To date, few functional markers that are commonly expressed on both human and murine HSCs have been reported. Thus, ESAM is important not only as a marker, but also for determining the identity of HSCs across species.

Recently CD166, a member of the immunoglobulin superfamily, was reported as a universal marker for murine and human HSCs [23]. This molecule was also expressed on endosteal niche components and immature osteoblasts, which exhibit high HSC-supporting activity [24,25]. Reciprocal transplantation of CD166-deficient HSCs into wild-type mice and of wild-type HSCs into CD166-deficient mice indicated that homophilic CD166–CD166 interaction likely mediates the HSC–niche association and affects HSC maintenance. Given that ESAM is also a member of the immunoglobulin superfamily with homophilic interacting features, it may also be involved in such HSC–niche interactions. However, because ESAM is an endothelial lineage-related antigen, this molecule may mediate the interaction between HSCs and endothelial cells. Indeed, irradiation or 5-fluorouracil treatment upregulates ESAM expression on murine HSCs, resulting in an intimate association between ESAM^{High} HSCs and the vasculature of BM [12]. In this context, both irradiation and 5-fluorouracil decreased CD166 expression on HSCs [23]. Therefore, we propose that these two markers are important for HSC maintenance, but they do not necessarily have overlapping functions.

Although it is widely accepted that human HSCs reside in the CD34⁺CD38[−] fraction, recent studies reported that a CD34⁺CD38^{Low} fraction also contains human hematopoietic repopulating cells [26,27]. Our flow cytometry analysis revealed high ESAM expression on CD34⁺CD38^{Low} cells (Fig. 1A, right). Thus, high ESAM expression may enrich HSCs regardless of CD38 expression. Furthermore, long-term reconstitution activity has also been observed in the CD34[−] population [28–31]. Previous studies by Sonoda et al. have verified that cells negative for 18 lineage-related antigens and positive for CD133 effectively enrich SCID-repopulating HSC activity in human CB CD34[−] populations [32–34]. Because ESAM expression covers a broader range of CB cells that are negative for lineage-related antigens compared with CD34 expression (T. Yokota, unpublished observation), ESAM, as well as CD133, may be useful as a positive marker for the CD34[−] HSCs. Future studies will provide a more accurate marker set for estimating HSC numbers in human CB.

Human CB contains various types of progenitor cells that are useful for tissue regenerative therapy [35]. For example, the CB CD34⁺ fraction can contribute to vasculature development in zebrafish [36]. Furthermore, the administration of human CB CD34⁺ cells ameliorates neurologic damage after brain injury in rodents, presumably by stimulating angiogenesis [37]. Thus, CB CD34⁺

cells likely contain a substantial amount of angiopoietic progenitors. In the present study, we identified a new subpopulation of CB CD34⁺CD38[−] cells that express high levels of ESAM; these cells are apparently related to the endothelial lineage. The “ESAM^{Bright}” phenotype may be useful for enriching angiopoietic progenitors. However, thorough examination of their differentiation potential *in vitro* and *in vivo* is required to evaluate this possibility.

Because high ESAM expression was related to the totipotent nature and robust growth potential of normal HSCs, we expected that ESAM would be expressed in many leukemia cell lines. However, all tested myeloid and lymphoid cell lines were ESAM negative. In contrast, erythroid and megakaryocyte cell lines exhibited robust ESAM expression. Given that ESAM is indispensable for normal erythropoiesis and thrombocyte function [12,38], the molecule may be intrinsically involved in the erythrocyte and megakaryocyte lineages.

We observed that the human AML cell line KG-1a, which is normally ESAM[−], became ESAM⁺ concomitant with engraftment in immunodeficient mice. Although we cannot exclude the possibility that serial xenogeneic transplantation induced some transformation in KG-1a cells, there are other biological explanations for this phenotypic change. For example, ESAM may be expressed in a rare subpopulation of KG-1a cells that constitute the leukemic stem cell (LSC) compartment. In this context, a recent approach to identifying LSC markers revealed ESAM in the plasma membrane proteome of LSC-enriched fractions obtained from myeloid leukemia patients with poor prognosis [39]. Furthermore, such rare ESAM⁺ KG-1a cells may acquire higher proliferative activity *in vivo* by taking advantage of the interaction with the leukemia-supporting microenvironment in the BM [40]. Indeed, ESAM⁺ rKG-1a cells exhibited higher engraftment capacity compared with ESAM[−] rKG-1a cells. In this case, targeting the leukemia–microenvironment interaction via ESAM may be a therapeutic strategy, as ESAM may confer a more aggressive clinical course in some types of leukemia. As illustrated in Figure 6, the pattern of ESAM expression varied significantly among primary AML cases. Although we observed no correlation between ESAM expression and clinical features at this stage, further studies and longer observation periods may reveal the true clinical value of ESAM in the diagnosis and treatment of acute leukemia.

In summary, we found that ESAM is a robust functional marker of HSCs in humans as well as in mice. Additionally, ESAM is expressed on some types of human leukemia cells and may be useful for lineage determination and as a prognostic indicator. The cross-species properties of ESAM expression will accelerate the translation of findings in animal models to studies in humans; this will ultimately increase our understanding of the identity and biology of human HSCs and leukemia cells.

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Author contributions

TI and TY designed and performed the research, analyzed the data, and wrote the article. YK designed the research, analyzed the data, and wrote the article. HT established rKG-1a cells. MI, TS, and YS designed and performed the microarray analyses and analyzed the data. YD, TU, AT, YH, SE, HS, and KO designed the research and analyzed the data.

Conflict of interest disclosure

The authors declare no conflict of interest.

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Supplementary methods

Mononuclear cell preparation

Cord blood was diluted 1:1 with Dulbecco's phosphate-buffered saline without calcium and magnesium (PBS–, Nacalai Tesque, Kyoto, Japan) and overlain on an equivalent volume of Ficoll–Paque PLUS (GE Healthcare Life Sciences, Little Chalfont, UK). Cells were centrifuged for 25 min at 20°C at 400g. MNCs were isolated and washed twice with RPMI-1640 medium (Nacalai Tesque).

Flow cytometry analysis and cell sorting

Allophycocyanin (APC)-conjugated anti-CD34 (HPCA-2); fluorescein isothiocyanate (FITC)-conjugated anti-CD38 (HIT2) and CD45 (HI30); and phycoerythrin (PE)-conjugated anti-CD4 (SK3), CD8 (SK1), CD13 (WM15), and CD33 (P67-6) monoclonal antibodies (mAbs) were purchased from BD Biosciences (San Diego, CA). PE-conjugated anti-CD45 (HI30), CD90/Thy1 (5E10), CD133 (AC133), Flk1/VEGFR-2 (7D4-6), VE-cadherin (BV9), and CD19 (HIB19) were purchased from BioLegend (San Diego, CA). ESAM mAb (Clone 408519) was purchased from R&D Systems (Minneapolis, MN) and biotinylated using EZ-Link Sulfo-NHS-Biotin and biotinylation kits (Thermo Scientific, Waltham, MA). Streptavidin conjugated with PE–Texas Red (Life Technologies, Carlsbad, CA), PE, and APC (BD Biosciences) were used for visualization of biotinylated antibodies. Dead cells were excluded by staining with 7-aminoactinomycin D (7-AAD, Calbiochem, San Diego, CA). Cells were washed and then resuspended in PBS– containing 3% fetal bovine serum (MP Biomedicals, Irvine, CA). Cells were analyzed using FACSaria or FACSCanto II (BD Biosciences, Franklin Lakes, NJ). In some experiments, CD34⁺CD38[–]ESAM^{–/Low}, CD34⁺CD38[–]ESAM^{High}, and CD34⁺CD38[–]ESAM^{Bright} cells were sorted using a FACSaria and used in subsequent experiments. The FACS data was analyzed using FlowJo software (Tree Star, San Carlos, CA).

Colony-forming cell assays

For myeloid colony formation, cells were cultured for 14 days in MethoCult H4434 (StemCell Technologies, Vancouver, Canada). Subsequently, colonies were enumerated and classified as colony-forming unit granulocyte–macrophage (CFU-GM), burst-forming unit erythroid (BFU-E),

or CFU-Mix according to their shape and color, as observed under an inverted microscope. All cultures were incubated at 37°C in a humidified chamber with 5% CO₂.

Limiting dilution assays

Cord blood CD34⁺CD38[–]ESAM^{High} or CD34⁺CD38[–]ESAM^{Bright} cells were plated at various concentrations from 1 to 100 cells/well. Each well contained 200 μ L MSC growth medium with 10 ng/mL stem cell factor (SCF) and 5 ng/mL Flt-3 ligand. Half of the culture medium was replaced with fresh medium containing the same cytokines twice per week. After 28 days of culture, wells containing cells that had expanded were scored.

SCID repopulation assays

NOD/Shi-scid, IL-2 γ null (NOG) mice were purchased from the Central Institute for Experimental Animals (Kawasaki, Kanagawa, Japan). Sorted CB cells (CD34⁺CD38[–]ESAM^{–/Low}, ESAM^{High}, or ESAM^{Bright}) were injected via the tail vein after whole-body irradiation at a dose of 1.8 Gy. After 3 months, BM cells were obtained and analyzed by flow cytometry. Whole BM MNCs (2×10^6 cells/mouse) from primary recipients were injected into secondary recipients via the tail vein after whole-body irradiation. After 4 months, BM cells were obtained and analyzed by flow cytometry. Animal studies were performed with the approval of the institutional review board of Osaka University.

Microarray

Total RNAs were extracted from CD34⁺CD38[–]ESAM^{High} or ESAM^{Bright} cells of collagenase-treated BM using an RNeasy Micro Kit (Qiagen, Hilden, Germany). RNA samples were subjected to gene expression analysis using a SurePrint G3 Human GE 8 $\times$ 60K v2 Microarray (Agilent Technologies, Santa Clara, CA).

KG-1a mouse leukemia model

KG-1a cells derived from a human acute myeloid leukemia (AML) sample were transplanted into the tail vein of 2-Gy-irradiated NOD/SCID mice. Ten weeks after transplantation, reconstituted KG-1a (rKG-1a) cells were obtained from the BM of recipients. The rKG-1a cells were cultured in vitro for 3 weeks and then transplanted into the tail vein of 2-Gy-irradiated NOD/SCID mice.